

- 1 The photographs show two species of potatoes: sweet potatoes and white potatoes.



(Source: © min hee park/Alamy Stock Photo)

sweet potatoes



(Source: © Pakula Piotr/Shutterstock)

white potatoes

The tissue in the tubers contains molecules that can be used to produce new plants.

- (a) The tissues of the tubers contain starch molecules.

Explain why starch is an energy storage molecule.

(2)

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(b) Describe an experiment to compare the water potential of the tissue from these two types of tuber.

(5)

(c) (i) State **one** abiotic and **one** biotic variable, other than the independent variable, that could affect this experiment.

(2)

Abiotic variable

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Biotic variable

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(ii) Choose **one** of the variables you have identified in (i). Explain how this variable could be controlled. Describe what effect it could have on the results if it is not controlled.

(2)

Variable

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How this variable is controlled

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Effect it could have on the results if it is not controlled

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(d) Explain the factors that may affect the water potential of potato cells.

(3)

(Total for Question 1 = 14 marks)
